A Random Forest-Based Algorithm to Distinguish *Ulva prolifera* and *Sargassum* From Multispectral Satellite Images

Yanfang Xiao, Rongjie Liu, Keunyong Kim, Jie Zhang, and Tingwei Cui

***Abstract*— In 2017, large-scale macroalgae blooms with different dominant species of *Ulva prolifera* and *Sargassum* occurred concurrently in the Yellow and East China Seas, which poses a challenge to the cognition and control of macroalgae disaster. Therefore, it is necessary to develop an algorithm to distinguish *U. prolifera* and *Sargassum* from satellite images. In this study, the spectral difference between *U. prolifera* and *Sargassum* and the capability of several multispectral satellite missions to distinguish them is first analyzed. The results show that the reflectance peak in visible wavelength is always in ∼550 nm for *U. prolifera* whether it is floating in clear open water or turbid nearshore water. However, the reflectance of *Sargassum* floating in clear and turbid water shows totally different characteristics, because most of *Sargassum* body is submerged in the water and the observed *Sargassum* reflectance is seriously affected by water reflectance. Compared with Landsat 8 Operational Land Imager (OLI), HuanJing-1, Charge-Coupled Devices (HJ-1 CCD), Aqua Moderate-resolution Imaging Spectroradiometer (MODIS), and Sentinel 2 Multi-Spectral Instrument (MSI), GaoFen-1, Wide Field of View (GF-1 WFV) can preferably capture the spectral difference between *U. prolifera* and *Sargassum*. Based on the spectral difference analysis, we propose a random forest-based algorithm to distinguish *U. prolifera* and *Sargassum* from GF-1 WFV images with an overall accuracy of 97.6% except when *U. prolifera* and *Sargassum* mix together. The algorithm is more robust than the existing ones as it allowed more *Sargassum* samples from different ocean regions to be used in the training; in addition, it avoids negative effects caused by the selection of a threshold. The proposed algorithm is proved effective in distinguishing *U. prolifera* and *Sargassum* in the Yellow and East China Seas in May and June 2017 and in detecting *Sargassum* in the Atlantic Ocean. Thus, this method can be used in researches including floating macroalgae traceability and competition and succession between different macroalgae species in different regions of the ocean with similar environments.**

***Index Terms*— Machine learning algorithms, multispectral imaging, remote monitoring, tide.**

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I. Introduction

MARINE floating macroalgae bloom has become a constant occurrence for some regions worldwide. Green tide with *Ulva prolifera* as the main species has occurred in the Yellow Sea of China every summer since 2007; it has become a major local marine ecological disaster [1]–[6]. In recent years, the prevalence of another macroalgae bloom caused by *Sargassum* has been increasing in the coastal waters of Zhejiang and Jiangsu provinces of China [7], [8]. In December 2016, large amounts of floating *Sargassum* drifted to the laver culture areas near the coast of Jiangsu province, which resulted in a large economic loss of up to 500 million yuan (State Oceanic Administration, 2018). In May 2017, *U. prolifera* and *Sargassum* concurrently bloomed in the Yellow and East China Seas and drifted northward as a whole under the action of wind and current [8], [9]. The Gulf of Mexico and the Atlantic are high incidence areas of *Sargassum* [10]–[12]. Researchers have observed a large recurrent *Sargassum* belt that has been extending from West Africa to the Gulf of Mexico since 2011. The biomass of *Sargassum* was more than 20 million metric tons [13]–[15].

Accurate information of macroalgae distribution and their temporal changes is useful for research, such as determining the origins of the macroalgae and understanding their role in local biogeochemistry and carbon cycling, and management, such as rental equipment for beach cleaning and providing guidance for field surveys. Satellite remote sensing data are considered as the most important tool for monitoring marine floating macroalgae. However, the concurrent appearance of various species has increased the difficulty of monitoring macroalgae worldwide, including regions such as the Yellow and East China Seas from space.

Studies have mainly focused on finding ways to detect and monitor a single macroalgae species at a time, *U. prolifera* or *Sargassum* [13], [14], [16]–[25]. Almost all detection methods have been proposed owing to their high reflectance in green and near-infrared bands and high absorption in the red band [14], [16], [23], [26]. These methods perform well when a single macroalgae species occurs in a given region at a given time. However, when two macroalgae species occur in

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the same region at the same time, though these methods can easily detect the floating macroalgae, distinguishing between *U. prolifera* and *Sargassum* is difficult.

Several researchers were involved in distinguishing *U. prolifera* and *Sargassum*. Hu *et al.* [13] analyzed the spectral characteristics in a field by measuring reflectance of *Sargassum*, *Trichodesmium*, *U. prolifera*, garbage, and emulsified oil, and built a stepwise rule using several indices to fingerprint *Sargassum* pixels from the planned hyperspectral satellite missions. Hyperspectral satellite missions are advantageous for distinguishing *U. prolifera* and *Sargassum*; however, available hyperspectral images are not enough, although PRISMA (Italian Space Agency) is already operating, EnMAP mission will start soon and ESA Copernicus CHIME is also under consideration. Using multispectral satellite images, Zhang [27] constructed a green algae index (GAI) to distinguish *U. prolifera* and *Sargassum* in the Yellow and East China Seas based on spectral characteristics analysis of these two species. Min *et al.* [28] proposed the slope of red-green (SRG) index based on the reflectance slopes at green and red bands after Rayleigh correction to distinguish *U. prolifera* and *Sargassum*. However, GAI and SRG are only applied in the offshore regions of the Yellow Sea and do not analyze the applicability for nearshore regions; the indices are not evaluated quantitatively. In addition, threshold setting is required when GAI and SRG are used, which needs to be adjusted according to the quality of satellite image.

In this study, we first analyzed the spectral characteristics of *U. prolifera* and *Sargassum* using *in situ* hyperspectral reflectance, 6S atmospheric radiative transfer model, and the multispectral satellite remote sensing data from multiple satellite sensors. Second, a random forest-based algorithm was proposed to distinguish *U. prolifera* and *Sargassum* using numerous samples collected from the Yellow Sea, the East China Sea, and the Atlantic Ocean. Third, the random forest-based algorithm was applied to obtain the distributions of *U. prolifera* and *Sargassum* in the Yellow and East China Seas in May and June 2017 and applied for *Sargassum* detection in the Atlantic Ocean. Finally, we discussed multispectral requirements and the influence of mixed pixels on distinguishing *U. prolifera* and *Sargassum*.

II. Data and Methods

A. In Situ Hyperspectral Reflectance of U. Prolifera and Sargassum

U. Prolifera samples were collected from the coastal water off Qingdao on July 10, 2015. The samples were placed evenly in a water tank filled with seawater and surrounded by black plastic membrane to measure the reflectance of *U. prolifera*. The reflectance was measured outside using a FieldSpec4 ASD spectrometer with 1-nm resolution from 350 to 2500 nm. The spectrometer detector which had a field-of-view angle of 15° was set approximately 25 cm above the seawater surface and was oriented downward toward the *U. prolifera* [22]. The algae reflectance was averaged from ten measurements as the detector aimed at different positions on the surface of the tank.

Sargassum samples were collected on February 11, 2015, and the reflectance was examined in the laboratory using

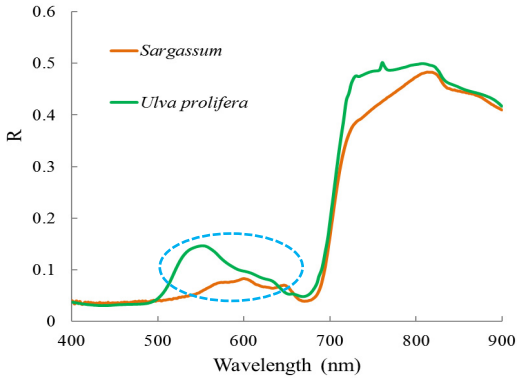


Fig. 1. *In situ* hyperspectral surface reflectance of *Ulva prolifera* (green line) and *Sargassum* (orange line). The peak value of reflectance (blue dotted line) in visible wavelength is approximately 550 nm for *U. prolifera* and approximately 600 nm for *Sargassum*.

TriOS-RAMSES in the observation range of 320–950 nm with 3.3-nm intervals. The samples were evenly covered in a glassware blackened all around and then the water was poured into the container to fill the space in *Sargassum* samples. The measurement was performed at a relative azimuth angle of 135° away from the sun light to minimize the effect of glint, and the sensor was set on the algae surface at a 30° angle of incidence [28], [29]. The algae radiance was measured ten times as the detector aimed at different spots on the surface of the container, and the measured reflectance was averaged as the reflectance of *Sargassum*. The reflectance can be converted from radiance as follows under the assumption that the algae samples are Lambertian:

$$R(\lambda) = \frac{L_{\text{alage}}(\lambda) \times \rho_{\text{r}}(\lambda)}{L_{\text{p}}(\lambda)} \tag{1}$$

where $R(\lambda)$ is the surface reflectance of algae floating on seawater; $L_{\text{alage}}(\lambda)$ and $L_{\text{p}}(\lambda)$ are the radiance of algae and reflection standard board, respectively. $\rho_{\text{r}}(\lambda, \theta)$ is the reflectance of reflection standard board (white reflection standard board is used here and $\rho_{\text{r}}(\lambda, \theta)$ is $\sim$100%). The *in situ* hyperspectral surface reflectance of *U. prolifera* and *Sargassum* is shown in Fig. 1. Both *U. prolifera* and *Sargassum* have the strongest reflectance in the near-infrared wavelength similar to terrestrial vegetation. The reflectance peak in visible wavelength is approximately 550 and 600 nm for *U. prolifera* and *Sargassum*, respectively.

B. Satellite Sensors

Spectral differences between *U. prolifera* and *Sargassum* are first analyzed from different satellite sensor perspectives, including Landsat 8 Operational Land Imager (OLI), GaoFen-1, Wide Field of View (GF-1 WFV), HuanJing-1, Charge-Coupled Devices (HJ-1 CCD), Aqua Moderate-resolution Imaging Spectroradiometer (MODIS), and Sentinel 2 Multi-Spectral Instrument (MSI). The purpose is to find the appropriate satellite sensor to distinguish *U. prolifera* and *Sargassum*. These satellite sensors are commonly used in macroalgae monitoring [32]–[38]. Table I shows the band setting of these sensors in visible and near-infrared wavelength.

GF-1 WFV images are then used in the training, validation, and application of the proposed algorithm because it can capture the more significant spectral differences between *U.*

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TABLE I
INFORMATION OF THE USED SATELLITE SENSORS IN VISIBLE AND NEAR-INFRARED WAVELENGTH

	Landsat 8 OLI	GF-1 WFV	HJ-1 CCD	Sentinel 2 MSI	Aqua MODIS
Band setting	Band1 430-450 nm	Band1 450-520 nm	Band1 430-520 nm	Band1 430-458 nm	Band1 620-670 nm
	Band2 450-510 nm	Band2 520-590 nm	Band2 520-600 nm	Band2 458-523 nm	Band2 841-876 nm
	Band3 530-590 nm	Band3 630-690 nm	Band3 630-690 nm	Band3 542-578 nm	Band3 459-479 nm
	Band4 640-670 nm	Band4 770-890 nm	Band4 760-900 nm	Band4 650-680 nm	Band4 545-565 nm
	Band5 850-880 nm			Band5 697-713 nm	
Spatial resolution	30 m	16 m	30 m	10 m	250/500/1000 m
Swath	185 km	200 km	360 km	290 km	2330 km
Revisit time	16 days	4 days	2 days	5 days	1-2 days

TABLE II
GF-1 WFV IMAGES USED IN THIS ARTICLE

Date	Data	Position	Object
2013/06/13	GF1_WFV1_E119.9_N34.7_20130613_L1A0000028262	Yellow and East China Seas	*U. prolifera*
2013/06/13	GF1_WFV1_E120.2_N36.3_20130613_L1A0000028261	Yellow and East China Seas	*U. prolifera*
2015/05/25	GF1_WFV4_E121.2_N33.5_20150525_L1A0000826615	Yellow and East China Seas	*U. prolifera*
2015/05/25	GF1_WFV4_E121.7_N35.1_20150525_L1A0000826614	Yellow and East China Seas	*U. prolifera*
2015/06/14	GF1_WFV2_E120.6_N36.0_20150614_L1A0000861912	Yellow and East China Seas	*U. prolifera*
2015/06/14	GF1_WFV3_E122.6_N35.6_20150614_L1A0000861930	Yellow and East China Seas	*U. prolifera*
2015/07/08	GF1_WFV1_E122.0_N36.3_20150708_L1A0000906437	Yellow and East China Seas	*U. prolifera*
2016/05/19	GF1_WFV2_E122.0_N34.3_20160519_L1A0001592654	Yellow and East China Seas	*U. prolifera*
2016/12/02	GF1_WFV2_E121.0_N34.3_20161202_L1A0002007511	Yellow and East China Seas	*Sargassum*
2016/12/31	GF1_WFV3_E120.5_N32.3_20161231_L1A0002084095	Yellow and East China Seas	*Sargassum*
2016/12/31	GF1_WFV3_E121.0_N33.9_20161231_L1A0002084094	Yellow and East China Sea	*Sargassum*
2017/05/11	GF1_WFV4_E125.8_N31.7_20170511_L1A0002354181	Yellow and East China Seas	*Sargassum*
2017/05/18	GF1_WFV1_E124.1_N34.7_20170518_L1A0002363893	Yellow and East China Seas	*Sargassum*
2017/05/18	GF1_WFV1_E123.4_N31.3_20170518_L1A0002363896	Yellow and East China Seas	*Sargassum*
2017/05/18	GF1_WFV1_E123.7_N33.0_20170518_L1A0002363894	Yellow and East China Seas	*Sargassum*
2017/05/18	GF1_WFV1_E124.1_N34.7_20170518_L1A0002363893	Yellow and East China Seas	*Sargassum*
2017/05/18	GF1_WFV1_E124.5_N36.3_20170518_L1A0002363889	Yellow and East China Seas	*Sargassum*
2017/05/27	GF1_WFV1_E120.8_N33.0_20170527_L1A0002382235	Yellow and East China Seas	*U. prolifera & Sargassum*
2017/05/27	GF1_WFV1_E121.2_N34.6_20170527_L1A0002382240	Yellow and East China Seas	*U. prolifera & Sargassum*
2017/05/27	GF1_WFV2_E122.8_N32.6_20170527_L1A0002382237	Yellow and East China Seas	*U. prolifera & Sargassum*
2017/05/27	GF1_WFV2_E123.2_N34.3_20170527_L1A0002382228	Yellow and East China Seas	*U. prolifera & Sargassum*
2017/05/27	GF1_WFV3_E124.6_N32.3_20170527_L1A0002396873	Yellow and East China Seas	*U. prolifera & Sargassum*
2017/06/09	GF1_WFV4_E123.0_N33.5_20170609_L1A0002417234	Yellow and East China Seas	*Sargassum*
2017/06/26	GF1_WFV4_E121.7_N35.0_20170626_L1A0002444678	Yellow and East China Seas	*U. prolifera*
2017/06/26	GF1_WFV4_E122.3_N36.6_20170626_L1A0002444677	Yellow and East China Seas	*U. prolifera*
2017/07/21	GF1_WFV1_W65.2_N12.9_20170721_L1A0002500054	Atlantic Ocean	*Sargassum*
2017/07/28	GF1_WFV1_W81.2_N24.6_20190728_L1A0004147811	Atlantic Ocean	*Sargassum*
2017/07/29	GF1_WFV4_W59.1_N10.5_20170729_L1A0002514459	Atlantic Ocean	*Sargassum*
2017/07/29	GF1_WFV4_W59.2_N10.1_20170729_L1A0002514461	Atlantic Ocean	*Sargassum*
2017/07/29	GF1_WFV4_W80.3_N28.5_20190729_L1A0004148865	Atlantic Ocean	*Sargassum*

prolifera and *Sargassum*. The images showing in Table II were obtained during the bloom period of *U. prolifera* and *Sargassum* in the Yellow and East China Seas and the Atlantic from 2013 to 2017. For *U. prolifera*, there are ten images from the Yellow and East China Seas that include different growth stages and different seawater environments (nearshore turbid water and offshore clear water). For *Sargassum*, there are ten images from the Yellow and East China Seas obtained between

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December 2016 (nearshore turbid water) and May 2017 (offshore clear water), and five images from the Atlantic in July 2017 (offshore clear water). The macroalgae type of these images is given according to field observations and the published literatures [8], [14], [15], [22], [27]. As the proposed algorithm is tuned with the reflectance after Rayleigh correction, all images here are Rayleigh-corrected imageries.

C. Radiative Transfer Simulation

Whether *U. prolifera* and *Sargassum* can be distinguished from satellite images depends on if the spectral differences between these two species can be captured by satellite sensors. To identify the differences and find which satellite sensor can capture the relatively significant difference, the best way is to obtain the reflectance of *U. prolifera* and *Sargassum* observed from different satellite sensors synchronously or quasi-synchronously; however, it is almost impossible to achieve. Therefore, the simulated satellite reflectance from *in situ* ground reflectance by the 6S atmospheric radiative transfer model is used here to analyze the spectral differences between *U. prolifera* and *Sargassum* for different sensor perspectives.

The 6S radiative transfer model is developed by Laboratoire d'Optique Atmospherique, URA CNRS 713, Université des Sciences et Technologies de Lille, Villeneuve d'Ascq Cédex, France, and is as one of the standard radiative transfer models used for remote sensing research and the creation of operational products [61]. It can accurately simulate the radiative transfer of an optical electromagnetic signal through the atmosphere. The input parameters of 6S model include the atmospheric conditions, altitude of the sensor and target, wavelength and ground reflectance, and so on. The outputs are at-sensor reflectance or radiance and a number of other calculated atmospheric parameters [62].

Using the 6S radiative transfer model, the *in situ* hyperspectral surface reflectance (R) was first simulated to hyperspectral reflectance after Rayleigh correction R_{rc} and hyperspectral reflectance on top of atmosphere R_{toa}. Table III shows the value of the main input parameters of 6S radiative transfer model in the simulation processing. The selected aerosol type was maritime aerosol [22], [23]. The satellite zenith angle and solar angle were set as 0° and 30°, respectively. The aerosol optical thickness (AOT) values at 550 nm (τ_{550}) were equal to 0.3. Studies have shown that the AOT value at 550 nm across the Yellow and East China Seas was usually 0.2–0.4 on sunny days [37].

Using the spectral response functions of different satellite sensors including Landsat 8 OLI, GF-1 WFV, HJ-1 CCD, Aqua MODIS, and Sentinel 2 MSI, the hyperspectral R, R_{rc}, and R_{toa} of *U. prolifera* and *Sargassum* were resampled to the equivalent multispectral surface reflectance, the equivalent multispectral reflectance after Rayleigh correction, and the equivalent multispectral reflectance on top of atmosphere using the following formula:

$$R_{\mathrm{sensor}}(\lambda) = \frac{\int F_0(\lambda) S_i(\lambda) R(\lambda) d\lambda}{\int F_0(\lambda) S_i(\lambda) d\lambda} \tag{2}$$

TABLE III
MAIN INPUT PARAMETERS OF 6S RADIATIVE TRANSFER MODEL

Main input parameters	Value
solar zenith angle	30°
satellite zenith angle	0°
solar azimuth angle	180°
satellite azimuth angle	180°
atmospheric model	mid-latitude in summer
aerosol model	maritime aerosol
aerosol concentration	0.3
target altitude	0 km
sensor altitude	700 km

where $S_i(\lambda)$ is the spectral response function of the ith band of a satellite sensor. $F_0(\lambda)$ is the solar irradiance outside the atmosphere at the average Earth–Sun distance. $R(\lambda)$ is the hyperspectral reflectance of *U. prolifera* and *Sargassum*. $R_{\mathrm{sensor}}(\lambda)$ is the equivalent multispectral reflectance of satellite sensors.

D. Random Forest Model

The random forest classifier is an ensemble classifier proposed by Breiman and Cutler [38] and is a common classification method for object-oriented image analysis [38]–[43]. It uses a set of classification and regression trees (CARTs) to make a prediction. The trees are created by random drawing a subset of training samples referred to as in-bag samples. The remaining samples (referred to as out-of-bag samples) are used to estimate the results of random forest model. To classify a new object from an input vector, the input vector passes through every tree in the forest. Each tree gives a classification, implying that the tree voted for the most popular class. The classification with the most votes is chosen as the classification result of the input vector (Fig. 2). Random forest has become popular within the remote sensing community due to its high classification accuracy, the speed of processing, and insensitive to overfitting [44]–[48].

The number of classification trees to be generated (Ntree) and the number of variables selected for the best split (Mtry) are two parameters that need to be set for a random forest model. The Mtry parameter is determined by analyzing the spectral difference between these two types of floating macroalgae which is described in detail in Section III. The number of classification trees is experimentally set according to the accuracy snapshots of each additional five trees which is described in Section IV.

E. Previous Algorithms to Distinguish U. Prolifera and Sargassum

Based on the spectral characteristic analysis of *U. prolifera* and *Sargassum* from the Yellow Sea, Ruddick *et al.* [29] and Zhang *et al.* [30] proposed GAI and SRG indices to distinguish *U. prolifera* and *Sargassum*, respectively. GAI is constructed based on the baseline height of green band after atmospheric correction to distinguish *Sargassum* and *U. prolifera*. It can be calculated as follows. Generally, the GAI value of *U. prolifera*

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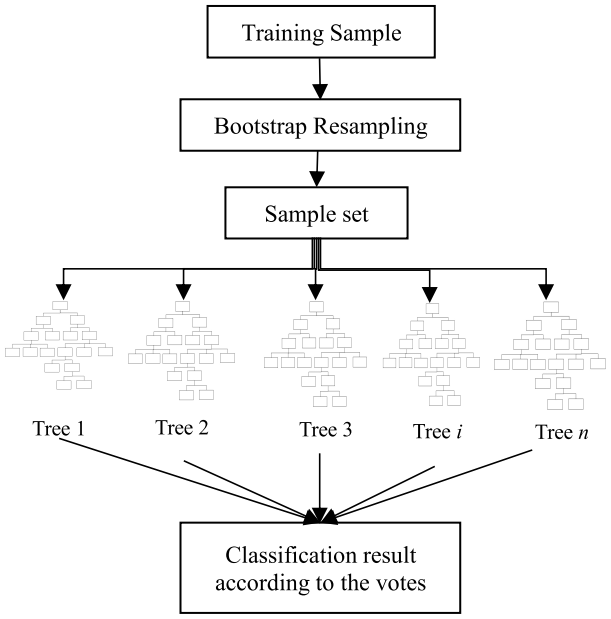


Fig. 2. Processing diagram of random forest model.

is higher than that of *Sargassum*; however, the threshold value to distinguish two algae is not fixed and needs to be adjusted according to a satellite image

$$\mathrm{GAI} = R_{_\mathrm{green}} - \frac{\lambda_{\mathrm{red}} - \lambda_{\mathrm{green}}}{\lambda_{\mathrm{red}} - \lambda_{\mathrm{blue}}} R_{_\mathrm{blue}} - \frac{\lambda_{\mathrm{green}} - \lambda_{\mathrm{blue}}}{\lambda_{\mathrm{red}} - \lambda_{\mathrm{blue}}} R_{_\mathrm{red}}. \tag{3}$$

SRG is built based on the R_{rc} slope between the green and red bands. It can be calculated as follows. To reduce the impact of seawater, R_{rc} used in the SRG is R_{rc} of the macroalgae after removing R_{rc} of nearby seawater from the entire reflectance. The SRG of *U. prolifera* is always less than zero and that of *Sargassum* is always larger than zero

$$\mathrm{SRG} = 1000 \times \frac{R_{\mathrm{rc_red}} - R_{\mathrm{rc_green}}}{\lambda_{\mathrm{red}} - \lambda_{\mathrm{green}}} \tag{4}$$

$$R_{\mathrm{rc}} = R_{\mathrm{rc_floating\ algae}} - R_{\mathrm{rc_nearby\ water}}. \tag{5}$$

III. Spectral Difference Between *U. Prolifera* and *Sargassum*

A. Difference in In Situ Hyperspectral Reflectance

The location of reflectance peak in visible wavelength provides a unique spectral difference to distinguish *U. prolifera* and *Sargassum*. For *U. prolifera*, the reflectance peak is approximately in 550 nm, whereas it is approximately 600 nm for *Sargassum*. The enhanced reflectance between 580 and 650 nm leads to the characteristic $R_{_\mathrm{red}} > R_{_\mathrm{green}}$ for *Sargassum*. Pigment type and content in the body of the macroalgae produces this spectral difference. Studies have shown that the main pigments in fresh *U. prolifera* were chlorophyll *a* and chlorophyll *b*, with a content ratio of approximately 3:1 [49]. Compared with *U. prolifera*, the main pigments in *Sargassum* are fucoxanthin, chlorophyll *a*, lutein, and zeaxanthin [51], all of which are carotenoids. Fig. 3 shows the specific absorption coefficient of chlorophyll $a + b$ and carotenoids from PROSPECT-5 leaf optical model [50], [61].

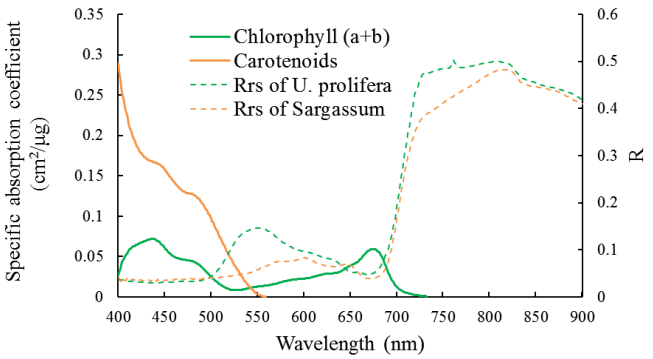


Fig. 3. Specific absorption coefficient of chlorophyll $(a + b)$ (green line) and total carotenoids (orange line).

The chlorophyll absorption peaks are at 440 and 680 nm, and the absorption at 500–600 nm is relatively low. A reflectance peak near 550 nm forms for *U. prolifera*. The carotenoids absorb at 400–550 nm and stop at around 550 nm [52]–[55]. It leads to the relatively high reflectance of ~600 nm for *Sargassum*.

B. Difference in the Reflectance of Satellite Sensors

As analyzed above, the location of the reflectance peak in the visible wavelength is the main hyperspectral difference between *U. prolifera* and *Sargassum*. Can the difference still be observed when the hyperspectral reflectance is resampled to the multispectral reflectance of satellite sensors? Which satellite sensor can capture the spectral difference?

Fig. 4 shows the simulated satellite multispectral reflectance (R, R_{rc}, and R_{toa}) of *U. prolifera* and *Sargassum* for several satellite sensors including GF-1 WFV, HJ-1 CCD, MODIS, Landsat 8 OLI, and Sentinel 2 MSI. For R, GF-1 WFV, HJ-1 CCD, and MODIS can capture the spectral difference between *U. prolifera* and *Sargassum*, which is $R_{_\mathrm{red}} > R_{_\mathrm{green}}$ for *Sargassum* and $R_{_\mathrm{green}} > R_{_\mathrm{red}}$ for *U. prolifera*. Landsat 8 OLI and Sentinel 2 cannot capture the spectral difference between *U. prolifera* and *Sargassum*. The spectral difference between *Sargassum* and *U. prolifera* in GF-1 WFV is the most significant. The ratio of $R_{_\mathrm{red}}$ and $R_{_\mathrm{green}}$ of *Sargassum* was 1.23, 1.07, and 1.03 for GF-1 WFV, HJ-1 CCD, and MODIS, respectively. The distinguishing ability of these satellite missions is related to the band setting and spectral response functions (Fig. 5). The strong response wavelength of green light for GF-1 WFV occurs in the range of 524–578 nm, which is entirely outside the reflectance peak of *Sargassum*. The strong response wavelength of red light occurs at 628–682 nm, which includes most of the region of the reflectance peak of *Sargassum* (565–650 nm). Therefore, the integration results show that $R_{_\mathrm{red}}$ is higher than $R_{_\mathrm{green}}$. Although, the strong response wavelength (624–662 nm) of the red light for MODIS is also in the reflectance peak region of *Sargassum* (565–650 nm), the integral value of $R_{_\mathrm{red}}$ is lower than that of GF-1 WFV because of the relatively low response value (average response in 624–662 nm is 82.6% for MODIS). In addition, the response wavelength of green light for MODIS is relatively narrow and close to the peak reflectance region of *Sargassum*, which resulted in $R_{_\mathrm{red}}$ of *Sargassum* being

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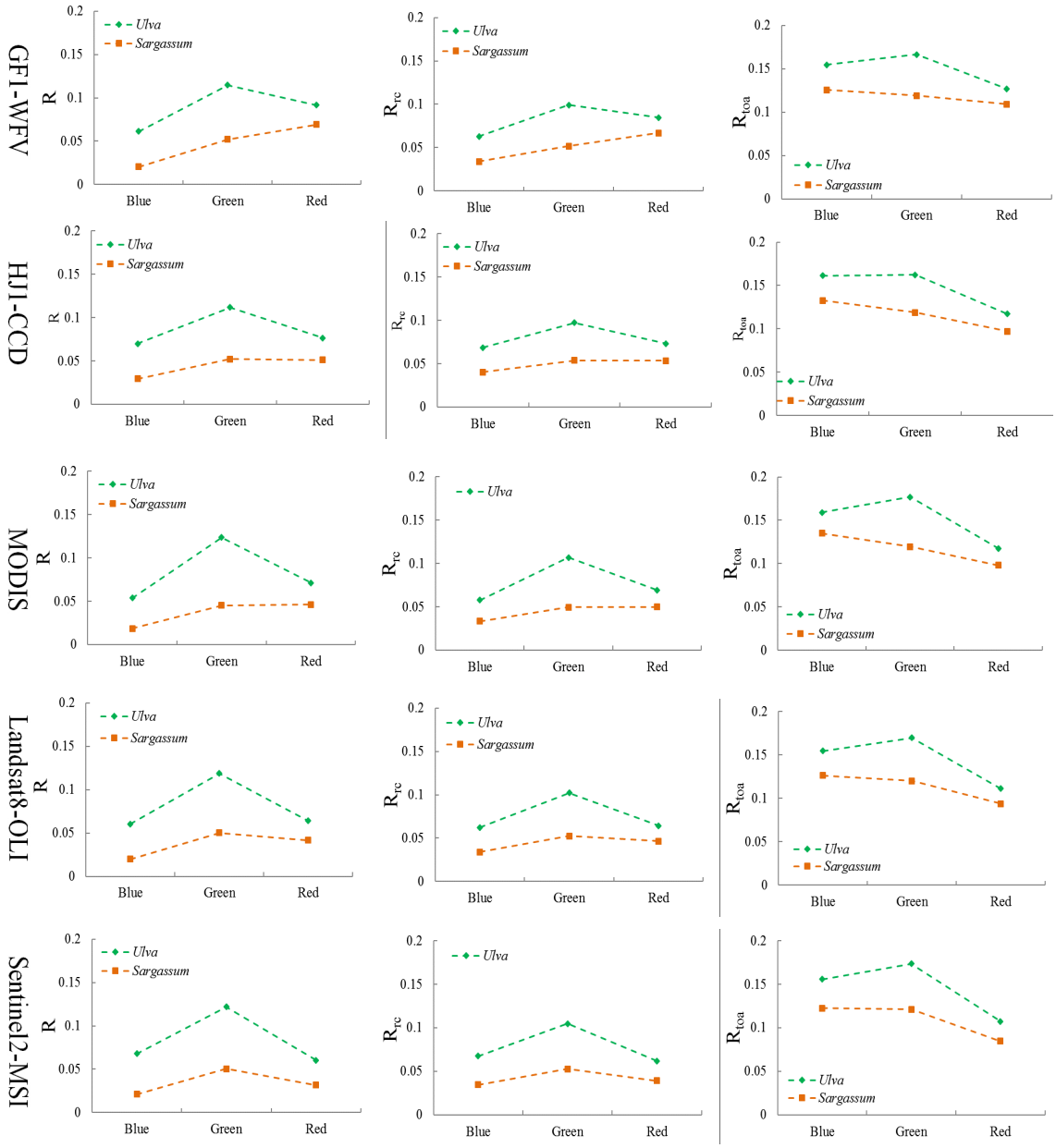


Fig. 4. Simulated R, R_{rc}, and R_{toa} of *Ulva prolifera* and *Sargassum* at Landsat 8 OLI, GF-1 WFV, HJ-1 CCD, MODIS, and Sentinel 2 MSI. *Sargassum* has the characteristic of R and R_{rc} in red band higher than R and R_{rc} in green band for GF-1 WFV, HJ-1 CCD, and MODIS, and the characteristic is most significant for GF-1 WFV (the ratio of $R_{_red}$ and $R_{_green}$ is 1.23, 1.07, and 1.03 for GF-1 WFV, HJ-1 CCD, and MODIS). The characteristic disappears for R_{toa}. *U. prolifera* has the characteristic of R and R_{rc} and R_{toa} in the red band less than R and R_{rc} and R_{toa} in the green band for all sensors.

slightly higher than R_{green}. The strong response wavelength of Landsat 8 OLI to green light is 536–587 nm, which is in the climbing region of *Sargassum* reflectance at 500–570 nm and the high value region between 570 and 650 nm. The strong response wavelength of Landsat 8 OLI to red light is 638–671 nm, most of which is in the descending region of *Sargassum* reflectance (650–672 nm). Sentinel 2 MSI is similar to Landsat OLI. The strong response wavelength of Sentinel 2 MSI to red light is 652–678 nm, which is in the strong absorption region of *Sargassum* (approximately 672 nm).

The performance for R_{rc} in these satellite sensors is similar to R. GF-1 WFV, HJ-1 CCD, and MODIS have the highest reflectance in the green band for *U. prolifera* and the highest reflectance in the red band for *Sargassum*. Among these missions, GF-1 WFV shows the most significant spectral difference between the two species.

R_{toa} showed insignificant spectral differences between the two species of all sensors used in this study.

GF-1 WFV is chosen to distinguish *U. prolifera* and *Sargassum* due to its most significant satellite spectral difference and the relatively high spatial resolution. Given that there is no reliable atmospheric correction method to derive R when the target is floating algae [24], the Rayleigh-corrected reflectance R_{rc} is used to build the distinguish algorithm here.

We collected GF-1 WFV images of *Sargassum* in Jiangsu (China) coastal waters and the Atlantic Ocean to verify the results from the simulated data. The images of the

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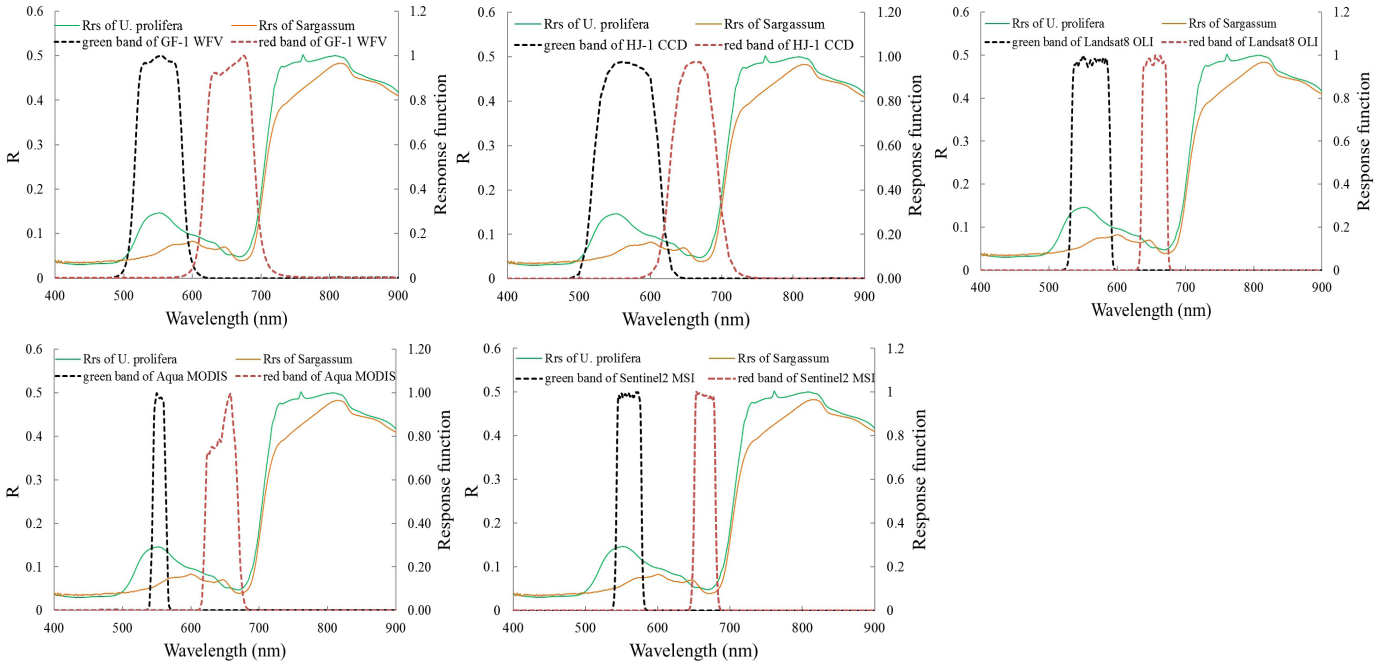


Fig. 5. Spectral response functions of the green and red bands for Landsat 8 OLI, GF-1 WFV, HJ-1 CCD, MODIS, and Sentinel 2 MSI.

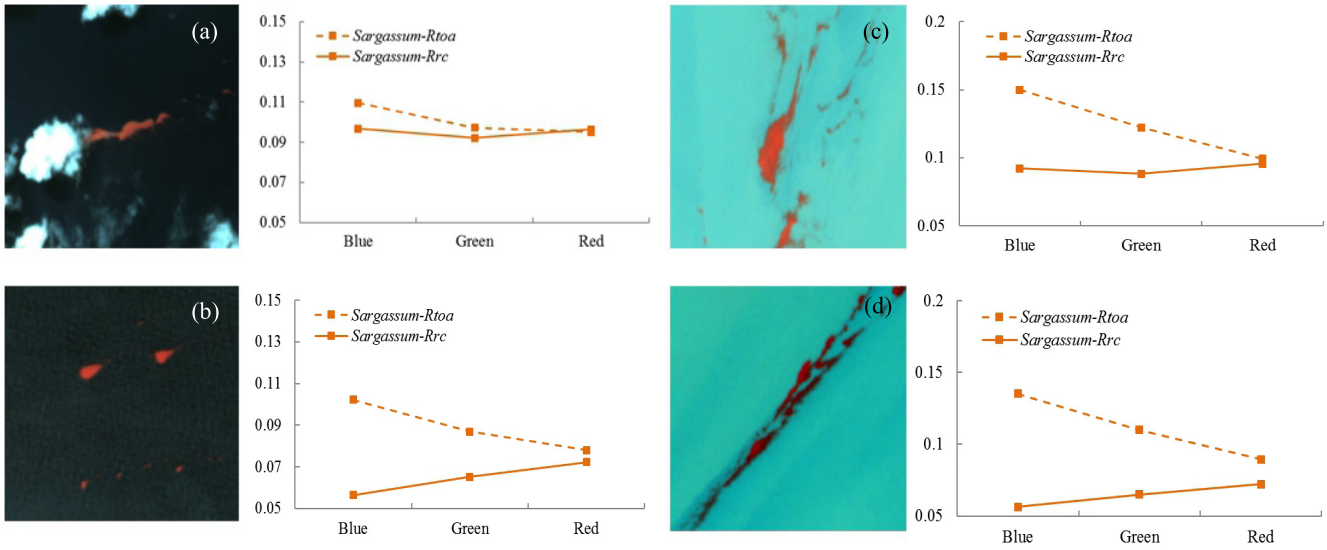


Fig. 6. GF-1 WFV images of *Sargassum* [combined by Band4 (red), Band3 (green), and Band2 (blue)] and its reflectance (R_{rc} and R_{toa}). (a) July 11, 2017, Atlantic Ocean (13.47° N, 65.43° W). (b) July 29, 2017, Atlantic Ocean (10.42° N, 61.97° W). (c) December 31, 2016, coastal waters of Jiangsu province (33.28° N, 121.42° E). (d) December 31, 2016, coastal waters of Jiangsu province (33.99° N, 121.62° E). *Sargassum* in Atlantic Ocean and the coastal waters of Jiangsu province has the characteristic of $R_{rc_red} > R_{rc_green}$ and similar to the simulated results.

Atlantic Ocean were acquired in July 2017, when *Sargassum* had the largest coverage [14], [15]. The images of Jiangsu coastal waters were acquired in December 2016, when a large-scale *Sargassum* bloom invaded the laver culture area (State Oceanic Administration, 2018). The results show that R_{rc} of *Sargassum* in the Atlantic Ocean and Jiangsu coastal waters of GF-1 WFV images have similar spectral characteristics of $R_{rc_red} > R_{rc_green}$ with the simulated results (Fig. 6).

In May 2017, a large-scale *Sargassum* bloom occurred in the Yellow and East China Seas [8], [9]. However, *Sargassum* in this period has different spectral characteristics, as shown in Fig. 7. R_{rc_blue} is the highest, followed by R_{rc_green} and R_{rc_red}. This spectral difference of *Sargassum* from different places and times is possibly caused by the pigment content and the water environment that *Sargassum* is floating on. This is further analyzed in Section V.

R_{rc} of *U. prolifera* under different growth stages (early/vigorous growth stage) and different water environment (clear/turbid water) was extracted from GF-1 WFV images (Fig. 8). The results show that the R_{rc} peak of *U. prolifera* always occurs in the green band, whether in the early or vigorous growth stage, and whether it is floating on clear or turbid water.

It proves that the spectral differences in the actual GF-1 WFV images between *U. prolifera* and *Sargassum* are also mainly in the visible bands after Rayleigh correction.

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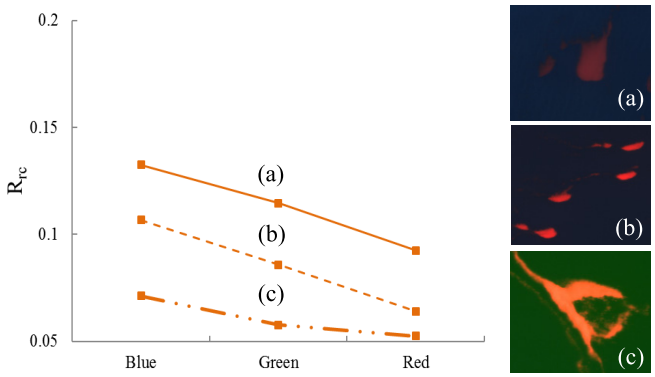


Fig. 7. GF-1 WFV images of *Sargassum* and its R_{rc} in the Yellow and East China Seas in May 2017. (a) May 6, 2017 (31.85° N, 124.46° E). (b) May 10, 2017 (31.59° N, 124.57° E). (c) May 18, 2017 (30.67° N, 126.49° E). *Sargassum* had the highest R_{rc} in the blue band which may have been caused by the pigment content and the water on which the *Sargassum* was floating.

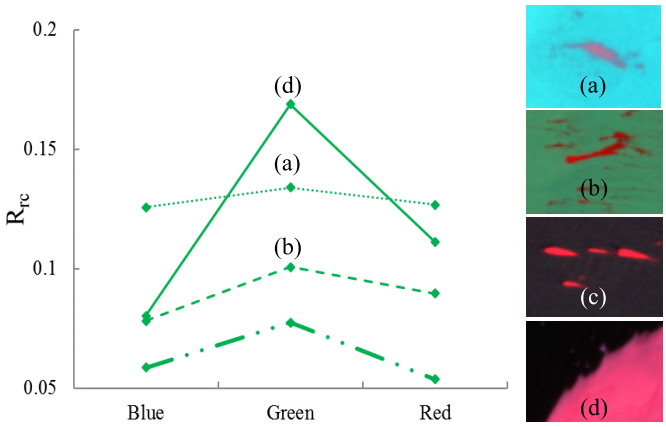


Fig. 8. R_{rc} of *Ulva prolifera* under different growth stages and different water environments. (a) Early growth stage in extremely turbid water (May 11, 2016, 33.44° N, 121.43° E). (b) Early growth stage in turbid water (May 19, 2016, 32.91° N, 121.32° E). (c) Early growth stage in clear water (May 19, 2016, 34.48° N, 121.71° E). (d) Vigorous growth stage in clear water (July 8, 2015, 36.07° N, 121.73° E). The R_{rc} peak of *U. prolifera* always occurs in the green band, regardless of whether it is in the early or vigorous growth stage and whether it is floating on clear or turbid water.

U. prolifera has the highest Rayleigh-corrected reflectance in green band, that is, it has the spectral characteristics of $R_{rc_green} > R_{rc_blue}$ and $R_{rc_green} > R_{rc_red}$. The spectral characteristics of *Sargassum* are either $R_{rc_red} > R_{rc_green}$ or $R_{rc_blue} > R_{rc_green} > R_{rc_red}$.

IV. Random Forest-Based Algorithm to Distinguish *U. Prolifera* and *Sargassum*

A. Samples and Training

We extracted approximately 10 700 *Sargassum*, 13 000 *U. prolifera*, and 25 000 seawater samples from 21 GF-1 WFV images to train the random forest-based algorithm. According to the spectral difference analysis between seawater, *U. prolifera*, and *Sargassum*, the Mtry parameters are set as R_{rc_blue}, R_{rc_green}, R_{rc_red}, R_{rc_NIR}, $EVI_{_Rrc}$, $(R_{rc_green} - R_{rc_blue})/(R_{rc_green} + R_{rc_blue})$, $(R_{rc_green} - R_{rc_red})/(R_{rc_green} + R_{rc_red})$, and $GAI_{_Rrc}$. R_{rc} is the Rayleigh-corrected reflectance. $EVI_{_Rrc}$ is the enhanced vegetation index which improves vegetation monitoring by reducing the background influences from the atmosphere and canopy and improves the sensitivity of the estimations in high biomass regions [64], [65]. Here, it is used to distinguish water and floating microalgae. $(R_{rc_green} - R_{rc_blue})/(R_{rc_green} + R_{rc_blue})$ and $(R_{rc_green} - R_{rc_red})/(R_{rc_green} + R_{rc_red})$ are two new indices built according to the spectral difference between *U. prolifera* and *Sargassum* analyzed above. $(R_{rc_green} - R_{rc_blue})/(R_{rc_green} + R_{rc_blue})$, $(R_{rc_green} - R_{rc_red})/(R_{rc_green} + R_{rc_red})$, and GAI are used to distinguish *U. prolifera* and *Sargassum*. Table IV shows the value range of those parameters for *U. prolifera*, *Sargassum*, and seawater. The number of classification trees is experimentally set as 100 according to the accuracy snapshots of each additional five trees (Fig. 9).

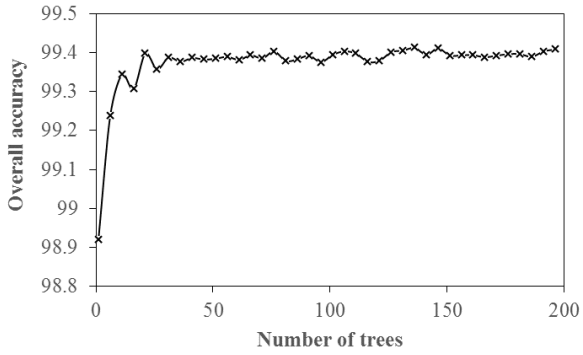


Fig. 9. Snapshots of over accuracies for distinguishing *Sargassum* and *U. prolifera* at five trees interval for random forest.

Fig. 10 shows a schematic flowchart of the entire distinguishing process. Two random forests are used to distinguish *U. prolifera* and *Sargassum*. The first random forest (RF1) is constructed using R_{rc_blue}, R_{rc_green}, R_{rc_red}, R_{rc_NIR}, and $EVI_{_Rrc}$ to distinguish seawater and floating macroalgae. The second random forest (RF2) is constructed using $(R_{rc_green} - R_{rc_blue})/(R_{rc_green} + R_{rc_blue})$, $(R_{rc_green} - R_{rc_red})/(R_{rc_green} + R_{rc_red})$, and $GAI_{_Rrc}$ index to divide the macroalgae into *U. prolifera* and *Sargassum*.

B. Algorithm Evaluation

To evaluate the proposed algorithm, 16 subgraphs with 400 × 400 pixels were extracted from other nine GF-1 WFV images of the Yellow and East China Seas and the Atlantic Ocean after data processing. Only one macroalgae species is present in each subgraph because the "real value" is needed to test the algorithm, and it is difficult and inaccurate to achieve a subgraph that consists of two macroalgae species that occur together. According to previous studies and field surveys, the *Sargassum* test samples were only taken from GF-1 WFV images of the Atlantic Ocean, Yellow Sea of China in December 2016, and East China Sea in early May 2017 [8], [14], [15], [18]. The test samples of *U. prolifera* were only taken from GF-1 WFV images of the Yellow Sea of China obtained in July 2012 and 2015. A confusion matrix is used to evaluate the accuracy of the distinguishing results (Table V), with an overall accuracy (OA) of 97.6% and kappa coefficient of 0.96. It should be noted that the samples used to evaluate the algorithm exclude the case that the two macroalgae species

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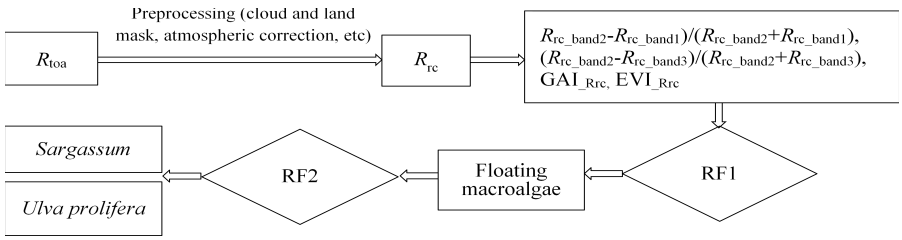


Fig. 10. Schematic flowchart shows the steps of distinguishing *Sargassum* and *U. prolifera* from satellite images.

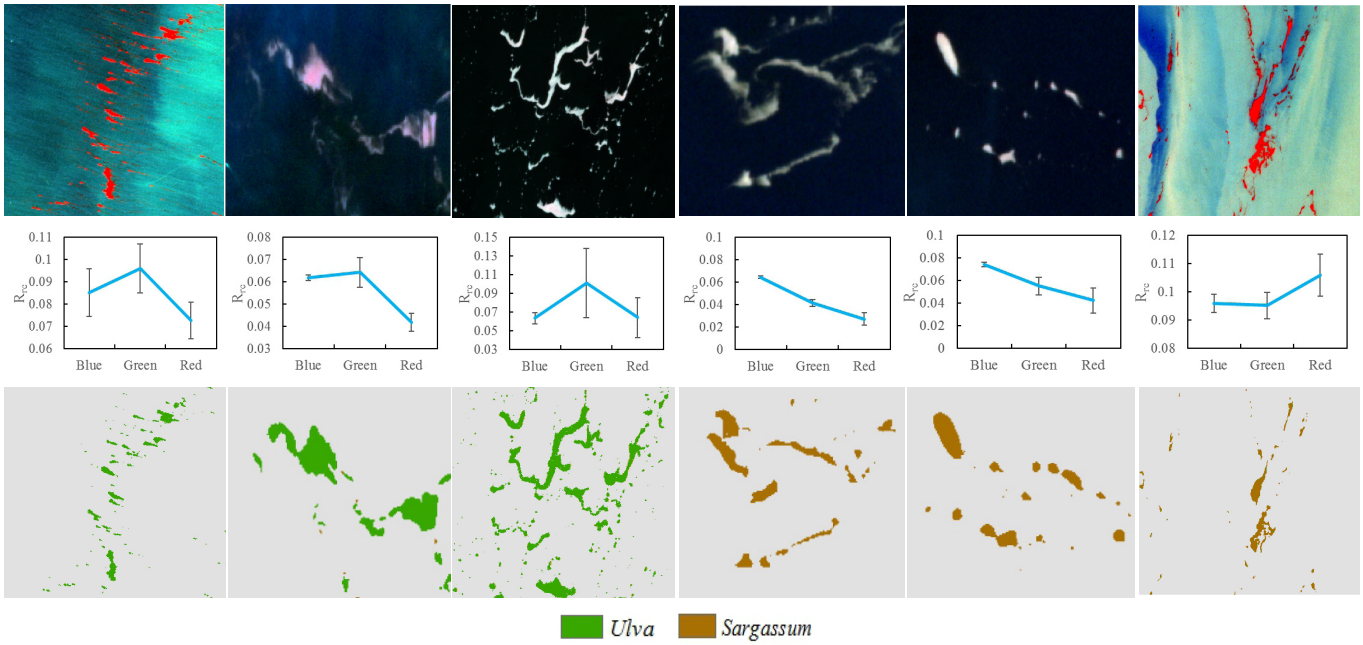


Fig. 11. Examples of distinguishing results of *Ulva prolifera* and *Sargassum* using the random forest-based algorithm. The top images are GF-1 WFV images (composed by band 4-near infrared response (NIR), band 3-red, and band 2-green) after Rayleigh correction. The middle line charts show the R_{rc} spectrum of the floating algae. The distinguishing results of *Ulva prolifera* and *Sargassum* using the random forest-based algorithm are at the bottom.

TABLE IV

R_{rc} OF *Ulva prolifera*, *Sargassum*, AND SEAWATER SAMPLES

	Ulva prolifera (13000 pixels)	*Sargassum* (10700 pixels)	Sea water (25000 pixels)
R_{rc_blue}	0.056~0.113	0.052~0.153	0.055~0.096
R_{rc_green}	0.062~0.197	0.046~0.134	0.044~0.107
R_{rc_red}	0.043~0.144	0.036~0.137	0.029~0.077
R_{rc_NIR}	0.180~0.457	0.094~0.305	0.016~0.053
EVI_{Rrc}	0.275~0.821	0.051~0.649	-0.163~0.015
(R_{rc_green}- R_{rc_blue})/(R_{rc_green}+ R_{rc_blue})	0.003~0.390	-0.180~0.117	-0.158~0.089
(R_{rc_green}- R_{rc_red})/(R_{rc_green}+ R_{rc_red})	0.001~0.288	-0.116~0.210	0.123~0.244
GAI_{Rrc}	0.001~0.089	-0.011~0.009	-0.005~0.022

mix. Fig. 11 shows examples of distinguishing results obtained using the random forest-based algorithm.

C. Comparison With Previous Algorithms

Here, the proposed random forest method is compared with GAI and SRG. Tables VI and VII show the confusion matrixes used to evaluate the accuracy GAI and SRG calculated from the same test samples as the random forest method. GAI is calculated from the reflectance after atmosphere correction referred to the process of Zhang [27]. The threshold of GAI is set as 0.01 according to estimate how well the distinguishing results of *U. prolifera* and *Sargassum* for the training samples used to produce the random forest model. For GAI, the OA is 86.8% and kappa coefficient is 0.72. The threshold of SRG is set as 0 according to Min *et al.* [28]. For SRG, the OA is 88.1% and kappa coefficient is 0.85. It should be noted that the first step to distinguish *U. prolifera* and *Sargassum* is get the floating algae for GAI and SRG. The floating algae was gotten by adjusting the value of normalized difference vegetation index (NDVI) until a better result is achieved in [27] and [28]. However, the floating algae here is from the result of RF1 process of the prosed random forest method in order to ensure the same samples for evaluation.

D. Algorithm Application

1) Distribution of Sargassum and U. Prolifera in the Yellow and East China Seas: Large-scale floating Sargassum and *U. prolifera* occurred in the Yellow and East China Seas from May and July 2017 [8], [25]. We analyzed the spatial and

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TABLE V
CONFUSION MATRIX OF THE RANDOM FOREST-BASED ALGORITHM

	*Ulva*_Random forest	*Sargassum*_ Random forest	Total samples	Recall
*Ulva*_labeled	52973	1806	54779	96.7%
*Sargassum*_labeled	113	25575	25688	99.6%
Total samples	53086	27381		
Precision	99.8%	93.4%		

TABLE VI
CONFUSION MATRIX OF GAI (THRESHOLD OF GAI = 0.01)

	*Ulva*_GAI	*Sargassum*_GAI	Total samples	Recall
*Ulva*_labeled	44188	10591	54779	80.7%
*Sargassum*_labeled	0	25688	25688	100%
Total samples	44188	36279		
Precision	100%	70.8%		

TABLE VII
CONFUSION MATRIX OF SRG (THRESHOLD OF SRG = 0)

	*Ulva*_SRG	*Sargassum*_ SRG	Total samples	Recall
*Ulva*_labeled	48430	6349	54779	88.4%
*Sargassum*_labeled	3214	22474	25688	87.0%
Total samples	51644	28823		
Precision	93.8%	78.0%		

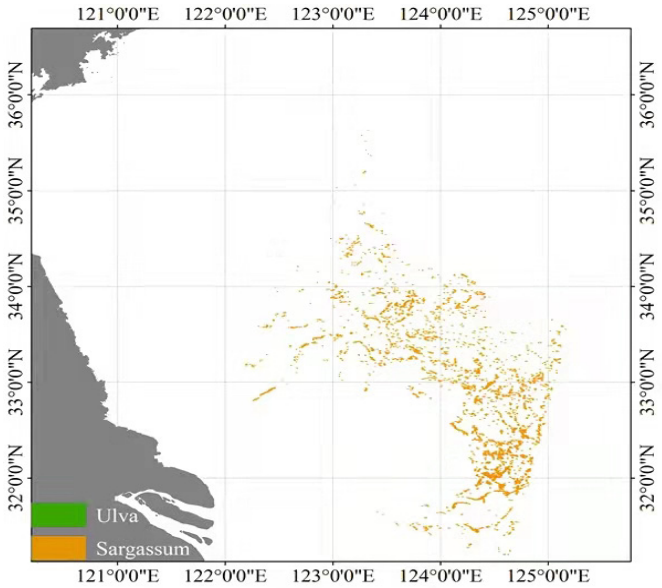


Fig. 12. Detection of *Sargassum* and *Ulva prolifera* in the Yellow and East China Seas (May 18, 2017). *Sargassum* is mainly distributed in the East Sea outside the Yangtze River Estuary.

temporal distribution of *Sargassum* and *U. prolifera* during this period using the random forest-based algorithm. In the middle of May (May 18, 2017) (Fig. 12), the main species of floating macroalgae was *Sargassum*, accounting for 99.9% of the area of total floating macroalgae. *Sargassum* mainly distributes in the East Sea outside the Yangtze River Estuary. At the end of May (May 27, 2017) (Fig. 13), the area ratio of *Sargassum* to *U. prolifera* was approximately 2:3. The floating *Sargassum* mainly appeared near 121° E–122° E, 34.5° N–35.5° N (Region 1) and 122° E–124° E, 33° N–34° N (Region 2). *U. prolifera* mainly appeared in Region 1.

Floating *Sargassum* disappeared at the end of July (June 26, 2017) and *U. prolifera* was the main macroalgae in the Yellow and East China Seas (Fig. 14). There was some uncertainty in the analysis of *U. prolifera* and *Sargassum* distribution in the Yellow and East China Seas from GF-1 WFV images because the images did not cover the whole sea area.

2) Application for Sargassum Detection in Atlantic Ocean: Large quantities of *Sargassum* were recently reported in the central Atlantic Ocean and Caribbean Sea [10], [13]–[15], [24], [55]. In the training of the proposed algorithm, samples from the Atlantic Ocean are used to analyze the application of the proposed algorithm in *Sargassum* detection in the Atlantic Ocean. In July 2019, a large-scale *Sargassum* occurred near Florida Strait. We obtained the GF-1 WFV images of this region on July 28 and extracted the floating macroalgae using the proposed algorithm. The algorithm performed well in distinguishing the floating macroalgae as *Sargassum* (Fig. 15). No quantitative evaluation of the proposed method has been carried out for the Atlantic Ocean because of the few effective GF-1 images.

V. DISCUSSION

A. Reasons Leading to Spectral Difference of Sargassum

The reflectance of *Sargassum* observed at different locations and times is diverse, as shown in Section III-B. Pigment in algae and the water environment that the algae is floating on are the main reasons that lead to diversity.

First, *Sargassum* from different places and times shows apparent color differences that are determined by pigment type and content. The field survey in 2017 revealed that the *Sargassum* collected in the east of 122° E was golden yellow, whereas *Sargassum* collected in the west of 122° E (120°30′ –121°30′ E) was brown. The change of macroalgae colors resulted from the change of pigment type and content in the algal body [8]. Zhang *et al.* [51] found that the pigment content of *Sargassum* floating offshore of China varied significantly. The chlorophyll *a* content of *Sargassum* floating nearby Dongji Island (122.4° E, 30.1° N) was 22.6 times that of *Sargassum* floating nearby Rizhao (119.6° E, 35.3° N).

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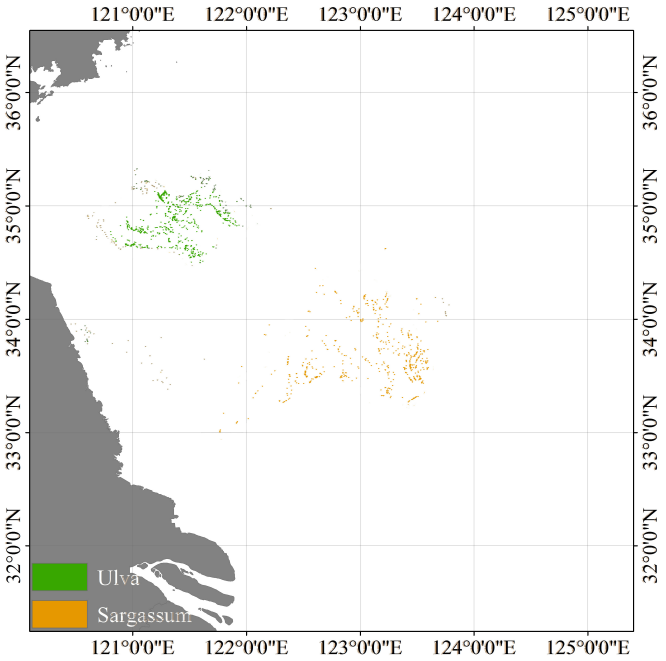


Fig. 13. Detection of *Sargassum* and *Ulva prolifera* in the Yellow and East China Seas (May 27, 2017). The floating *Sargassum* mainly appeared near 121° E–122° E, 34.5° N–35.5° N (Region 1) and 122° E–124° E, 33° N–34° N (Region 2). *U. prolifera* mainly appeared in Region 1.

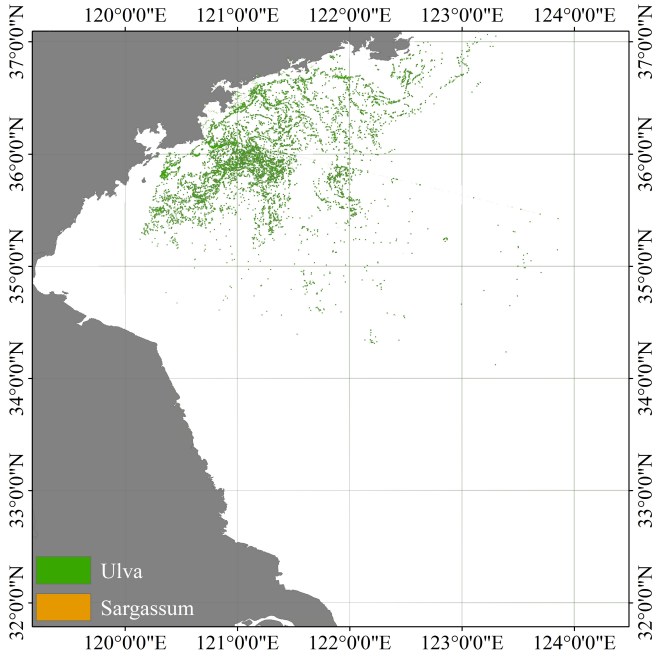


Fig. 14. Detection of *Sargassum* and *Ulva prolifera* in the Yellow and East China Seas (June 26, 2017). *Sargassum* disappeared and *U. prolifera* was the main macroalgae in Yellow and East China Seas.

Sargassum floating on the Yellow Sea had a higher content of fucoxanthin and xanthophylls compared with *Sargassum* floating on the East China Sea. Terasaki *et al.* [56] showed seasonal change in the content of fucoxanthin of *Sargassum* found in Japanese coastal waters. Endo *et al.* [57] analyzed the pigment changes of *Undaria pinnatifida* under different lighting and nutritional conditions and found that the fucoxanthin content significantly increased under low light and high nutrition, and that the chlorophyll *a* content increased with high nutrition. The difference in pigment type and the change in pigment content may lead to diverse spectral characteristics for *Sargassum*.

Second, the water environment that *Sargassum* is floating in is different. *U. prolifera* has a tubular hollow structure that fills the algae with gas. Therefore, most of the *U. prolifera* body is above the water [Fig. 16(a)]. Compared with *U. prolifera*, the body of *Sargassum* is larger and its floating ability only depends on the distribution of air sacs in the middle or top of the larger lateral branches. This causes most of the *Sargassum* body to be below water [Fig. 16(b)]. Thus, the *Sargassum* signals observed above are not entirely from *Sargassum*, and the signals consist of large amounts of water information. Fig. 16(c) shows R_{rc} of *Sargassum* floating on turbid water and clear water, and R_{rc} of nearby water taken from GF-1 WFV images. The water reflectance will greatly affect the spectral shape of *Sargassum* reflectance.

B. Size Distribution Statistics of U. Prolifera and Sargassum

Several researchers have proved that floating macroalgae patches can be very small. Xiao *et al.* [20] analyzed the distribution of *U. prolifera* in the Yellow Sea using HJ-1 CCD images (30-m spatial resolution). The result shows that approximately 96.9% of *U. prolifera* patches are <0.25 km^2 (equal to the area of one MODIS pixel with a 500-m spatial resolution). The area of each *Sargassum* patch in the Yellow and East China Seas was counted based on the monitoring results from GF-1 WFV images in May 18, 2017. Among the detected 67.314 *Sargassum* patches, the maximum area of the patches is only 5.2 km^2, and only 221 patches are larger than 0.25 km^2, which account for 0.33% of the total number of patches. It means that the *Sargassum* patches are smaller and it is very hard to find pure pixels in low-resolution satellite images.

Qi *et al.* [25] monitored the temporal and spatial distribution of *Sargassum* in the Yellow and East China Seas from March to June 2017 using MODIS images and alternative floating algae index (AFAI). The result showed that the area of floating macroalgae in the Yellow and East China Seas was approximately 580 km^2 on May 18, 2017. Kim *et al.* [58] monitored the areas of green and gold tide in the Yellow Sea from 2008 to 2017 using geostationary ocean color imager (GOCI) data. The gold tide area in the eastern Yellow Sea reached 1299.8 km^2 in May 26, 2017. Compared with previous studies, our result shows that the area of floating *Sargassum* in the East China Seas is approximately 443.3 km^2 in May 18, 2017, which is calculated by multiplying the number of the identified *Sargassum* pixels and the area of single pixel (256 m^2) of GF-1 WFV image. Notably, GF-1 WFV images on this day did not cover the total area of the Yellow and East China Seas. The difference in the area of floating *Sargassum* from different areas from other studies is possibly caused by the spatial resolution, time, and coverage of the satellite data used.

C. Spectral Requirements for Satellite Images to Distinguish U. Prolifera and Sargassum

The spectral analyses in 3.1 clearly show that the location of reflectance peak in visible wavelength can be used as the

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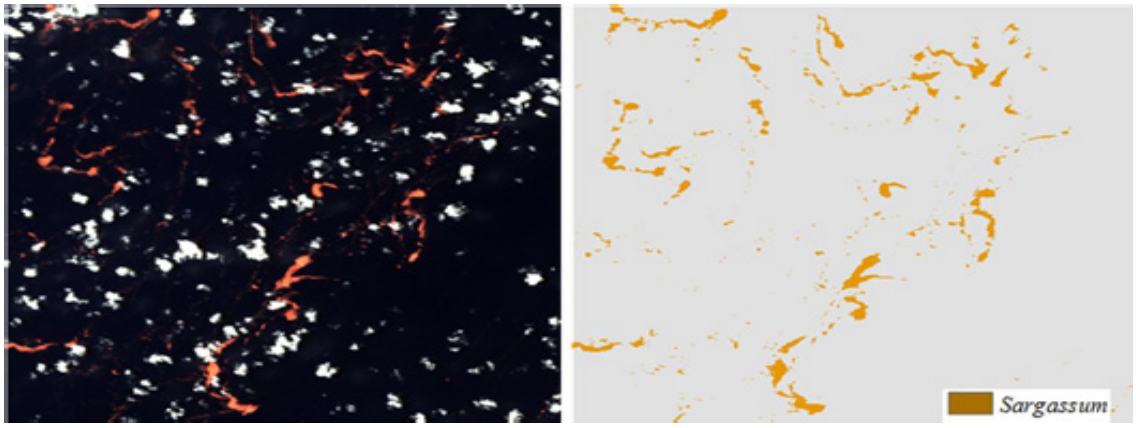


Fig. 15. Floating *Sargassum* in the Florida Strait. (Left) GF-1 WFV image (July 28, 2019, 24.02° N, 82.40° W, combined by Band4, Band3, and Band2). (Right) Detected floating *Sargassum* using the proposed algorithm.

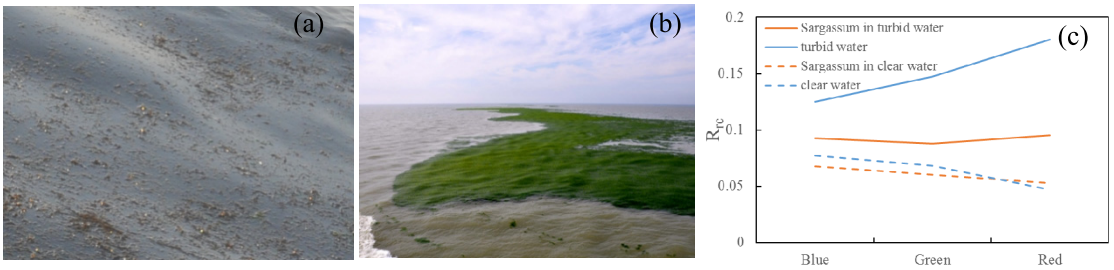


Fig. 16. Effect of seawater on the spectral characteristics of floating macroalgae. (a) Floating *Sargassum* from field observation (May 29, 2017). (b) Floating *Ulva prolifera* from http://www.dxy.cn/bbs/topic/33947804?from=recommend (May 20, 2016). (c) R_{rc} of seawater (blue line) and *Sargassum* (orange line) in the middle of the Yellow Sea from GF-1 WFV images.

criterion to distinguish *U. prolifera* and *Sargassum*. For *U. prolifera*, the reflectance peak is always approximately 550 nm, whereas it is approximately 600 nm for *Sargassum*. Such a reflectance characteristic has been reported in Hu *et al.* [12] and Dierssen *et al.* [59] from field measurements. Based on the reflectance characteristic, Hu *et al.* [12] identified the spectral regions and spectral bands to differentiate *Sargassum* from *Trichodesmium*, *Syringodium*, *U. prolifera*, garbage, and emulsified oil at different spectral resolutions (1, 5, and 10 nm), and proposed a stepwise rule using several indices to fingerprint *Sargassum* pixels for the planned hyperspectral satellite sensors. Without considering the influence of spatial resolution, the satellite images with high spectral resolution have a great advantage to distinguish *U. prolifera* and *Sargassum* in which the reflectance difference can be best retained. However, hyperspectral satellite missions are scarce and have a long revisit time, which greatly limits application in macroalgae discrimination.

Multispectral satellite missions are numerous and have relatively short revisit time compared with hyperspectral satellite missions. The analysis in this article additionally proves that some of the satellite sensors, such as GF-1 WFV and HJ-1 CCD, can distinguish *U. prolifera* and *Sargassum*. The spectral response of green and red bands (including the wavelength range and the response value corresponding to each band) is the determinant factor for multispectral satellite missions to distinguish these two species. We simulated the right band setting of green and red bands for multispectral satellite mission with different spectral resolutions using the *in situ* reflectance of *U. prolifera* and *Sargassum* showed in Fig. 1 by assuming that the spectral response function is 1 (Table VIII).

TABLE VIII

BAND SETTING OF GREEN AND RED BANDS FOR MULTISPECTRAL SATELLITE MISSION AT DIFFERENT WAVELENGTHS (20, 30, 40, 50, AND 60 nm). THE DISCRIMINABILITY WAS CALCULATED AS THE DIFFERENCE IN THE VALUE OF $R_{rc_green} - R_{rc_red}/R_{rc_green} + R_{rc_red}$ BETWEEN *U. prolifera* AND *Sargassum*

	Wavelength range (nm)				
	20-nm	30-nm	40-nm	50-nm	60-nm
Green band	523-543	520-550	515-555	509-559	504-564
Red band	640-660	636-666	629-669	618-668	603-663
Discriminability between *U. prolifera* and *Sargassum*	0.5016	0.4568	0.4146	0.3793	0.3573

To distinguish *U. prolifera* and *Sargassum*, the center of the green band needs to move from ∼555 to ∼535 nm to avoid the rising region of *Sargassum* reflectance from green to red light (545–565 nm). The center of the red band needs to move from ∼660 to ∼650 nm to include the peak region of *Sargassum* reflectance (565–650 nm). The discriminability between *U. prolifera* and *Sargassum* decreased by 30% when the resolution was changed from 20 to 60 nm, which means higher spectral resolution can achieve better discrimination results.

D. Effect of Mixed Pixels on the Discrimination Between U. Prolifera and Sargassum

Mixed pixels result in overestimation of floating macroalgae coverage when satellite images with moderate resolution

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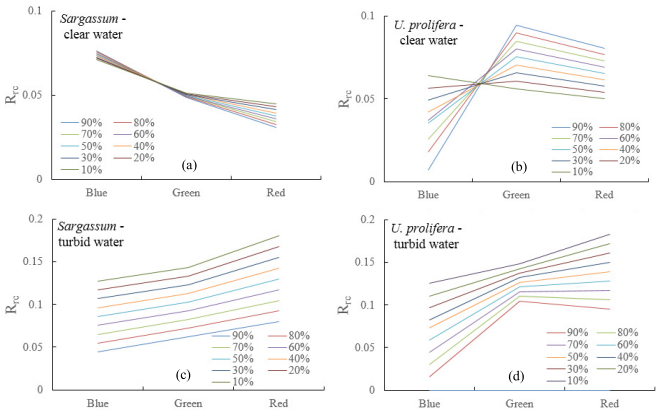


Fig. 17. (a) Mixed R_{rc} spectrum of *Sargassum* and clear water. (b) *Ulva prolifera* and clear water. (c) *Sargassum* and turbid water. (d) *U. prolifera* and turbid water. The proportion of water in the mixed pixels changes from 10% to 90%.



Fig. 18. Field survey of *Ulva prolifera* and *Sargassum* mixing together (June 5, 2017, Nantong coastal waters). The macroalgae with green color is *U. prolifera*, and the macroalgae with black color is *Sargassum*.

(tens of meters) are used [20], [21], [64]. Due to the influence of water in the mixed pixels, the spectral difference between *U. prolifera* and *Sargassum* may change, making it difficult to distinguish. Here, the mixed pixels were simulated by linear mixing *U. prolifera*/*Sargassum* and water endmembers in varying proportions ($R_{_mix} = f_{_water} \times R_{_water} + f_{_algae} \times R_{_algae}$, where $R_{_mix}$ is the reflectance of the mixed pixels, $R_{_water}$ is the reflectance of seawater endmember, $R_{_alage}$ is the reflectance of *U. prolifera*/*Sargassum*, and $f_{_water}$ and $f_{_algae}$ are the proportions of water and *U. prolifera*/*Sargassum* in the mixed pixels varying from 0.1 to 0.9, respectively), to determine the minimum percentage of macroalgae coverage in an image pixel that can retain the spectral difference. The reflectance spectra of *U. prolifera*/*Sargassum* and water endmembers were obtained from GF-1 WFV R_{rc} images of the Yellow Sea and the East China Sea. Two scenarios were simulated in this study. The first scenario simulated the mixed pixels in clear water, and the endmembers were obtained from the central Yellow and East China Seas. In this scenario, R_{rc} of the mixed *Sargassum* pixels retained the spectral characteristic of $R_{rc_blue} > R_{rc_green} > R_{rc_red}$ when the percentage coverage of *Sargassum* changed from 0.9 to 0.1 [Fig. 17(a)]. R_{rc} of the mixed *U. prolifera* pixels retained the spectral characteristics of the highest R_{rc_green} with at least 20% coverage of *U. prolifera* [Fig. 17(b)]. In other words, the minimum macroalgae proportion was 20% (80% water) for distinguishing the mixed *U. prolifera* and *Sargassum* pixels in clear water. The second scenario simulated the mixed pixels in turbid water, and the endmembers were obtained from the Subei shoal, which was considered the origin of *U. prolifera*. In this scenario, R_{rc} of the mixed *Sargassum* pixels retained the spectral characteristic of $R_{rc_red} > R_{rc_green}$ when the percentage coverage of *Sargassum* changed from 0.9 to 0.1 [Fig. 17(c)]; however, the spectral characteristic of *U. prolifera* was not retained when the coverage was less than 70% [Fig. 17(d)]. This indicates that the minimum macroalgae proportion was 70% (30% water) for distinguishing the mixed *U. prolifera* and *Sargassum* pixels in turbid water. In summary, the simulated results suggest that the mixed pixels may cause more interference with regard to adequate discrimination between *U. prolifera* and *Sargassum* in turbid water such as that in Subei shoal.

Our field survey in the Yellow Sea and the East China Sea led to the realization of mixing between *U. prolifera* and *Sargassum* (Fig. 18). When *U. prolifera* and *Sargassum* meet, *U. prolifera* mounts *Sargassum*, while *Sargassum* tears the patches of *U. prolifera* because of the differences in their morphological structure. For such a scenario, the proposed algorithm recognizes the mixed species as one macroalgae species based on the mixed reflectance spectra and cannot differentiate whether the pixels are mixed or pure. Therefore, the detection result of May 27, 2017 is characterized by some uncertainties. *U. prolifera* and *Sargassum* are proved to occur concurrently in the Yellow Sea; however, it proves difficult to ascertain the actual area occupied by these two species.

VI. Conclusion

A threshold-free method based on random forest model is proposed here to distinguish *U. prolifera* and *Sargassum* blooms mainly for the Yellow Sea and the East China Sea. The distinguishing method tuned with the Rayleigh-corrected reflectance from multispectral satellite sensor is an effective method to distinguish *U. prolifera* and *Sargassum* floating on different seawater environmental conditions. Data comparison showed that the proposed method is more advantageous than GAI and SRG because of better distinguishing results, insensitivity to seawater environmental conditions, no adjustment for thresholds, and improvement in the speed of processing. The proposed methods and training can be used in various research applications, including floating macroalgae traceability and competition and succession between different macroalgae species in different oceanic regions with similar environments. Note that model simulation showed that GF-1 WFV has better ability to distinguish *U. prolifera* and *Sargassum* and the proposed method is built for this satellite sensor. Although the result of model simulation has showed that other satellite sensors like Landsat 8 OLI, MODIS, and Sentinel 2 MSI have limitations in distinguishing *U. prolifera* and *Sargassum*, more work is needed based on not only model simulation but also the real images.

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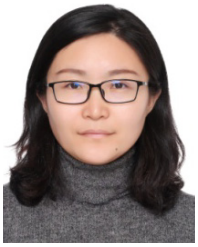
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